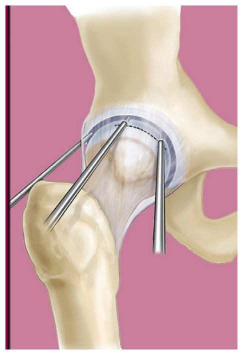

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Iliopsoas Pathology

Christian N. Anderson

The iliopsoas musculotendinous unit is a powerful hip flexor that is important for normal hip strength and function; however, disorders of the iliopsoas can be a significant source of pain and disability in the athletic population. These conditions include iliopsoas bursitis, tendonitis, impingement, and snapping; they have been shown to be the primary cause of chronic groin pain in 12% to 36% of athletes and are observed in 25% to 30% of athletes presenting with an acute groin injury.[1–4] Acute trauma may result in injury to the musculotendinous unit or avulsion fracture of the lesser trochanter in skeletally immature patients. Athletes with differing iliopsoas pathologies often present with a similar constellation of symptoms and findings on physical exam. Consequently it is important to develop an understanding of the normal anatomy and function of the iliopsoas as well as the pathophysiology of these disorders to accurately determine the diagnosis and formulate an appropriate treatment strategy.

Iliopsoas Anatomy and Function

The iliopsoas tendon-muscle complex is composed of the iliacus, psoas major, and psoas minor muscles (Fig. 82.1).[5] The psoas major is a long fusiform muscle that originates on the T12 to L5 vertebrae and intervertebral disks[5,6] and is innervated by the ventral rami of L1 to L4.[5,7] The iliacus is a triangular fan-shaped muscle composed of the medial, lateral, and ilioinfratrochanteric bundles[5,6,8–10]; it originates from the ilium and sacral ala and is innervated by the femoral nerve (L1-L2).[5,6,8,9] The psoas major and iliacus muscles converge at the level of the L5 to S2 vertebrae to form the iliopsoas muscle.[10] Prior to this convergence, the psoas major tendon originates above the level of the inguinal ligament from within the center of the psoas major muscle.[10] As the tendon courses distally, it rotates clockwise (right hip) and migrates posteriorly within the muscle, lying immediately anterior to the hip joint, and inserts on the lesser trochanter (Fig. 82.2).[10–12] The iliopsoas bursa is positioned between the musculotendinous unit and the bony surfaces of the pelvis and proximal femur. It typically extends from the iliopectineal eminence to the lower portion of the femoral head, with an average length of 5 to 6 cm and width of 3 cm.[10] The psoas minor is a long slender muscle that originates from the vertebral bodies of T12 and L1 and is present in only 60% to 65% of individuals.[13,14] Distally it merges with the iliac fascia and psoas major tendon, and in 90% of specimens it has a firm bony attachment to the iliopectineal eminence.[14]

Significant variability in the iliopsoas musculotendinous unit has been reported in the literature.[6,10–12,15–17] In a cadaveric study, Tatu and colleagues[10] reported the presence of two tendinous structures—the psoas major and iliacus. The medial iliacus muscle bundle was shown to insert onto the iliacus tendon, which progressively converges with the larger and more medial psoas major tendon.[10] The lateral muscle bundle of the iliacus courses distally, without any tendinous attachments, and inserts on the anterior surface of the lesser trochanter and infratrochanteric ridge.[10] These findings were corroborated in a study by Guillin and colleagues[9] using ultrasound (US) to map the iliopsoas anatomy. Conversely, in a study utilizing magnetic resonance imaging (MRI) with cadaveric correlation, Polster and coworkers[15] noted the medial iliacus bundle merged directly into the psoas major tendon, whereas the medialmost fibers of the lateral iliacus bundle inserted on a distinct thin intramuscular tendon. Philippon et al.[6] examined 53 fresh frozen cadavers and demonstrated, at the level of the hip joint, the prevalence of a single-, double-, and triple-banded iliopsoas tendon was noted 28.3%, 64.2%, and 7.5% of the time, respectively. However, in the pediatric population, the presence of two distinct tendons was observed in only 21% of patients undergoing MRI.[16] Gómez-Hoyos and coworkers observed a double (psoas and iliacus) tendinous footprint in 70% and a single tendon in 30% of specimens, which inserted on the anteromedial tip of the lesser trochanter, occupying 19% of its total surface.[11] Conversely, in a separate study by Philippon and coworkers,[12] the iliopsoas insertion was described as having an inverted teardrop shape that occupied the entire posterior surface of the lesser trochanter. Although controversy exists regarding the number of tendons, the relative contributions of the different muscle fibers to each tendon, and the location of the insertion point on the lesser trochanter, the current literature challenges historical descriptions of a single common conjoint tendon.

Reports of communication of the iliopsoas bursa with the hip joint through a congenital defect between the iliofemoral and pubofemoral ligaments are also variable. Tatu and colleagues[10] reported no communication in 14 cadaveric dissections; however, others have observed a direct communication between the joint and bursa in 15% of patients.[18] It is important to consider this communication during diagnostic injections, as the anesthetic material can move between the intra-articular and bursal compartments, thus confounding the results of the test.

Abstract

Disorders of the iliopsoas can be a significant source of groin pain in the athletic population. Commonly described pathologic conditions include iliopsoas bursitis, tendonitis, impingement, and snapping. The first line of treatment for iliopsoas disorders is typically a conservative program, including activity modification, physical therapy, nonsteroidal antiinflammatory drugs, and corticosteroid injections. Surgical treatment can be considered if the patient fails conservative measures and typically involves arthroscopic lengthening the musculotendinous unit and treatment of concomitant intra-articular pathology. Tendon release has been described at three locations—in the central compartment, in the peripheral compartment, and at the lesser trochanter—with similar outcomes observed between the techniques. Although results are generally favorable, transient hip flexor weakness is commonly observed but typically resolves by 3 to 6 months post surgery.

Keywords

coxa saltans interna
iliopsoas
iliopsoas bursitis
iliopsoas impingement
iliopsoas tendinitis
psoas
snapping hip

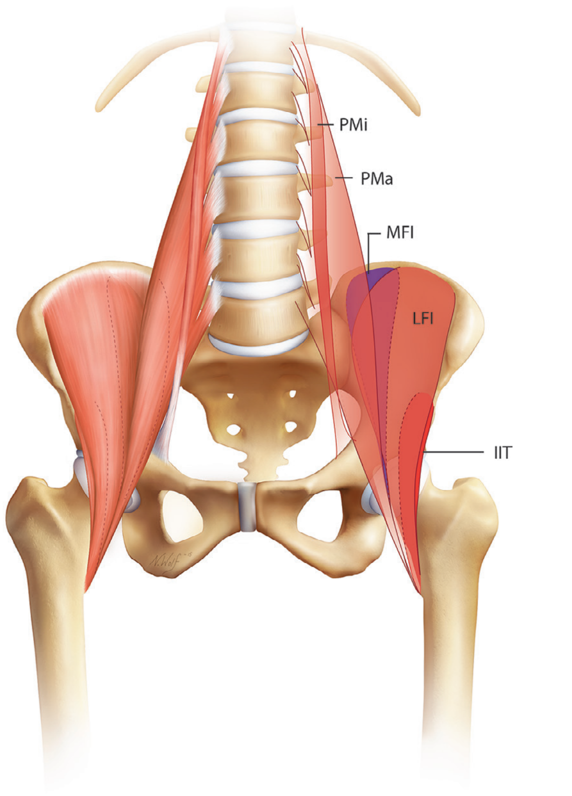

Fig. 82.1 Anteroposterior anatomy of the iliopsoas musculotendinous unit as described by Guillin et al[9] and Tatu et al.[10] *IIT*, Ilio-infratrochanteric muscle; *LFI*, lateral fibers of the iliacus; *MFI*, medial fibers of the iliacus; *PMa*, psoas major; *PMi*, psoas minor. (Copyright 2017, Nicole Wolf, all rights reserved.)

The iliopsoas unit functions primarily as a powerful hip flexor, but it also has important functions in femoral external rotation and with lateral bending, flexion, and balance of the trunk.[19–22] The iliacus and psoas major have been shown to have individual and task-specific activation patterns.[19,21] The iliacus is important for stabilizing the pelvis[17] and for early rapid hip flexion while running.[19] The psoas major is important for sitting in an erect position and for stability of the spine in the frontal plane.[21] Variable contributions of each muscle are observed during situps depending on the angle of hip flexion.[21] The exact function of the psoas minor has not been fully defined, but given its attachment to both the iliac fascia and bony pelvis, it has been hypothesized that it may assist in partially controlling the position and mechanical stability of the underlying iliopsoas as it crosses the femoral head.[14]

Iliopsoas Snapping

Iliopsoas snapping, also known as coxa saltans interna or internal hip snapping, is a disorder characterized by painful audible or palpable snapping of the iliopsoas during hip movement. In 1951, Nunziata and Blumenfled[23] first described the mechanism of internal coxa saltans as snapping of the iliopsoas tendon over the iliopectineal eminence of the pelvis. Since then, dynamic US has been used in several studies to confirm this mechanism as the primary source of iliopsoas snapping.[24–28] Most studies report a sudden "jerky" movement and audible or palpable snap of the iliopsoas over the iliopectineal eminence as the hip is brought from a position of flexion, abduction, and external rotation (FABER) to extension and neutral. Even so, more recent studies have demonstrated a sudden flipping of the psoas tendon over the iliacus muscle as the source of snapping (Fig. 82.3A–C).[9,24,29] In these studies, dynamic US revealed that as the hip was placed in the FABER position, the iliacus became interposed between the psoas tendon and the superior pubic rami (see Fig. 82.3A). As the hip was brought to neutral, part of the medial iliacus muscle became trapped as the tendon follows a reverse path to its original position (see Fig. 82.3B). The trapped iliacus is suddenly released, resulting in an audible snap of the tendon against the pubic bone (see Fig. 82.3C). Contrary to the original mechanism described by Nunziata and Blumenfeld, the iliopectineal eminence was medial to the psoas tendon and not involved with the observed snapping phenomenon.[29]

Although abnormal movement of the tendon over the bony pelvis is commonly described as the source of the snapping phenomenon, alternative mechanisms have been proposed. Several studies have proposed that soft tissue abnormalities—such as an accessory iliopsoas tendinous slip,[29] a paralabral cyst,[29] and or stenosing tenosynovitis—are the source of snapping.[30] Other authors have determined the iliopsoas snapping occurs over a bony prominence other than the pelvis, such as the lesser trochanters[31] or femoral head.[32] Overall, the exact mechanism of snapping remains controversial, and the lack of consensus regarding the mechanism supports the possibility of several potential etiologies for iliopsoas snapping.

Iliopsoas Bursitis and Tendonitis

Iliopsoas bursitis and tendonitis have been shown to be closely associated with the repetitive pathologic movement of the tendon observed in symptomatic coxa saltans interna.[27,31,33,34] The irregular movement of the tendon during the snapping phenomenon is thought to cause irritation and inflammation of the underlying bursa.[35–37] Even so, some studies have demonstrated no objective abnormality of the bursa in patients undergoing open surgery for symptomatic snapping.[31,38] Nevertheless, these conditions coexist so frequently that Johnson and coworkers[33] suggested they be considered a single entity referred to as "iliopsoas syndrome." Correspondingly, the diagnostic workup and treatment for these conditions are the same.

Iliopsoas Impingement

Iliopsoas impingement (IPI) is a pathomechanical process whereby an excessively tight iliopsoas tendon impinges upon the underlying acetabular labrum, resulting in labral injury.[39,40] This phenomenon was first described by Heyworth and colleagues in 2007[39] in a study (LOE IV) on revision hip arthroscopy; there they noted IPI and corresponding labral injury in 7 of 24 patients. After iliopsoas release at the level of the acetabulum, they noted that the tendon no longer impinged on the anterior labrum during hip extension.[39] Domb and coworkers (LOE IV)[40] further defined the pathophysiology of IPI in a series of patients with direct anterior labral tears at the 3 o'clock position (right hip) in the absence of bony abnormalities. They noted that the labral injury occurred directly beneath the iliopsoas tendon at the

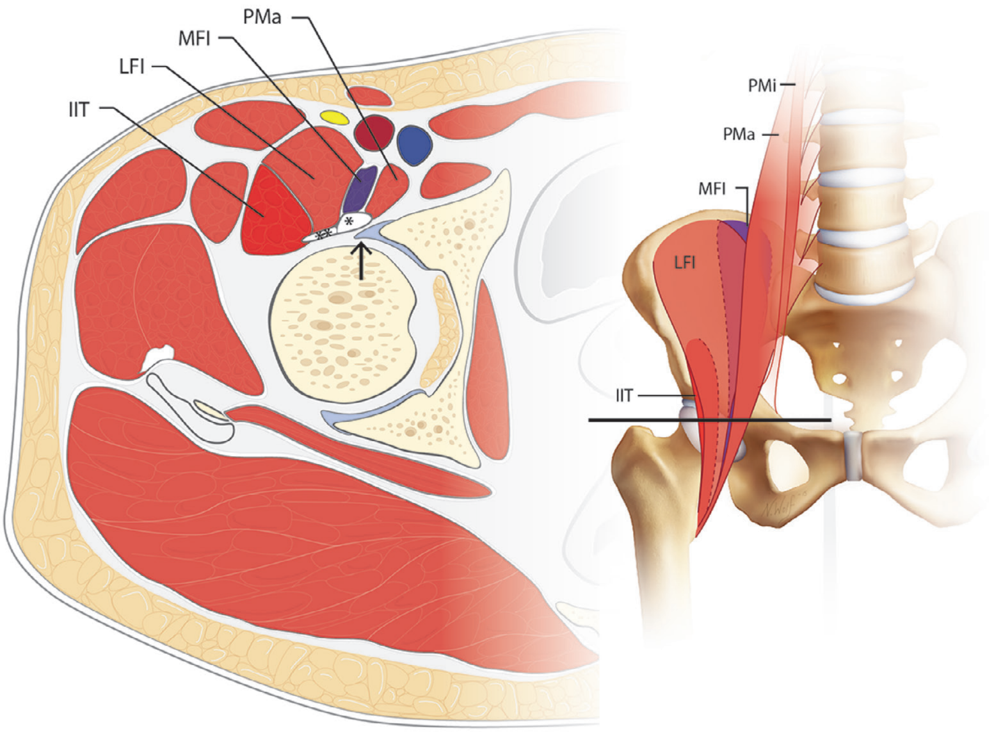

Fig. 82.2 Cross-sectional anatomy of the iliopsoas as described by Guillin et al[9] and Tatu et al.[10] The plane *(black line)* is through the hip joint, as demonstrated on the AP image. At this level, the iliacus *(**)* and psoas *(*)* tendons are posterior to the iliopsoas muscle bundles and anterior to the hip join and labrum *(arrow)*. *IIT*, Ilio-infratrochanteric muscle; *LFI*, lateral fibers of the iliacus; *MFI*, medial fibers of the iliacus; *PMa*, psoas major; *PMi*, psoas minor. (Copyright 2017, Nicole Wolf, all rights reserved.)

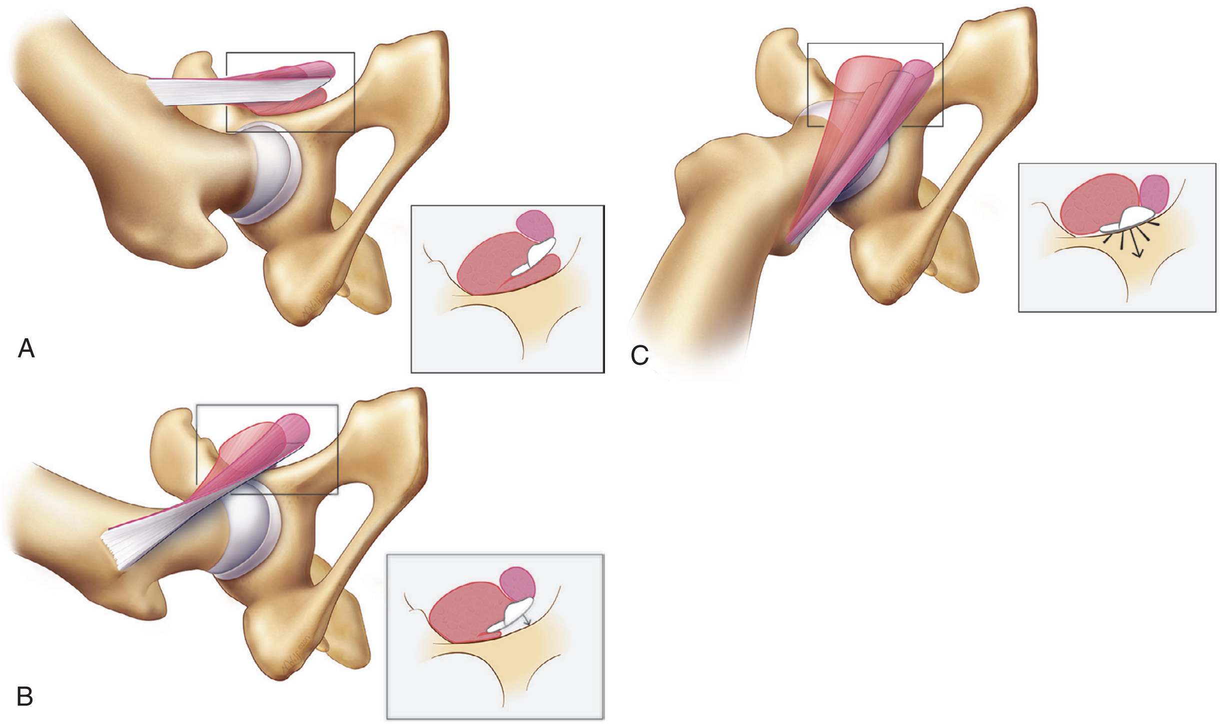

Fig. 82.3 Iliopsoas snapping mechanism as described by Deslandes et al.[29] With the hip flexed, abducted, and externally rotated, the iliacus muscle becomes interposed between the iliopsoas tendon and superior pubic ramus (A). When the hip is then brought to the neutral position, part of the iliacus muscle becomes trapped between the tendon and bone as the tendon follows a reverse path to its resting position on the pubic bone (B). As the hip is brought further into neutral, the trapped muscle suddenly releases, resulting in the tendon snapping against the adjacent pubic ramus (C). (Copyright 2017, Nicole Wolf, all rights reserved.)

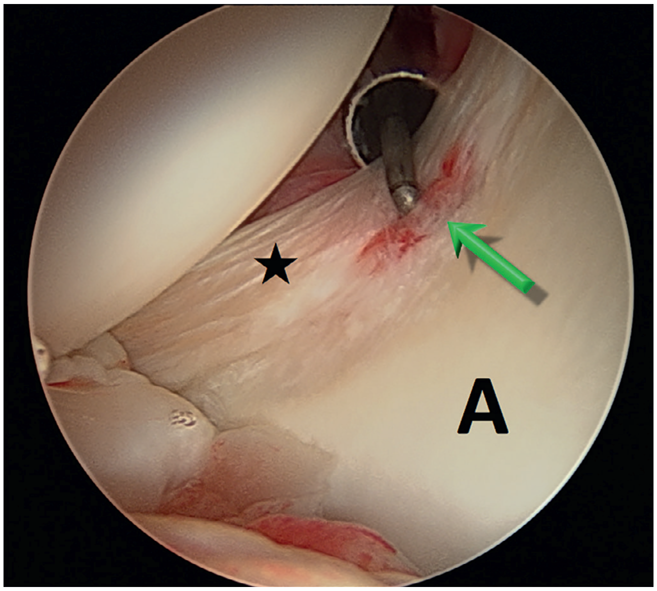

Fig. 82.4 Arthroscopic view from the posterolateral portal of a labral tear at the 3 o'clock position *(green arrow)* in a patient with iliopsoas impingement. The iliopsoas notch, anterior labrum *(star)*, and anterior acetabulum *(A)* are well visualized from this portal.

iliopsoas notch (Fig. 82.4), which significantly differs from the traditional 1 to 2 o'clock location observed in femoroacetabular impingement (FAI).[41,42] Pathologic findings at the time of arthroscopy included labral tearing, labral inflammation (referred to as the *IPI sign*), or adjacent tendinous inflammation and scarring of the tendon to the anterior capsule. The authors concluded that the labral injury was possibly the result of a tight iliopsoas impinging on the anterior labrum or a repetitive traction injury to the labrum from adherence of the tendon to the adjacent capsulolabral complex. Similar to the observations of Heyworth and colleagues, Domb and coworkers found that releasing the tendon decreased compression on the underlying labrum in all cases. A cadaveric study by Yoshio et al.[43] demonstrated that maximal pressure underneath the iliopsoas tendon occurs at the joint level during hip extension, supporting the possibility that excessive pathologic force from the iliopsoas possibly results in labral injury. Although the exact cause of the excessive tightening of the iliopsoas is unknown, Gómez-Hoyos and colleagues[44] observed that lesser trochanteric retroversion was significantly increased in patients with IPI compared with a control group. The authors postulated that because the iliopsoas tendon makes an obtuse angle as it crosses over the iliopectineal eminence, increasing retroversion of the lesser trochanter may elevate contact pressures underneath the tendon and contribute to impingement of the iliopsoas on the labrum.

HISTORY

Iliopsoas Snapping

The diagnosis of iliopsoas snapping begins with a thorough history. Patients often report painful snapping during sporting activities that require significant hip range of motion (ROM), such as dance, soccer, hockey, and football.[24,45] Symptoms can also occur during activities of daily living, such as climbing stairs or standing from a sitting position.[46] The snapping sensation is accompanied by groin pain, which may radiate into the thigh or top of the knee. A history of acute trauma has been associated with the development of snapping in up to 50% cases.[25] However, patients may also have preexisting asymptomatic snapping that becomes painful after repetitive training activities involving high hip flexion angles.[45] Symptomatic snapping is more common in females than males,[47] and although the true prevalence of this disorder is unknown, symptomatic snapping has been observed in up to 58% of elite ballet dancers.[24] Even so, the prevalence of asymptomatic snapping in the general population has been shown to be as high as 40%.[9] Therefore careful assessment is important in order to determine whether the snapping is symptomatic before proceeding with a treatment plan.

Iliopsoas Impingement

Iliopsoas impingement occurs most frequently in young active females, many who participate in regular sports.[40,48–50] Patients typically present with anterior groin pain that worsens with athletic activities and activities of daily living, such as active hip flexion, prolonged sitting, and getting out of a car.[40,48,49] Iliopsoas snapping is less commonly observed in IPI but has been reported in up to 17% of cases.[48,49]

PHYSICAL EXAM

Iliopsoas Snapping

Physical examination should include a complete musculoskeletal evaluation of the hip and focused specialty tests specific for the suspected diagnosis. The "active iliopsoas snapping test" is the most commonly described examination maneuver for detecting internal hip snapping. This test is performed by having the patient actively move the hip from the FABER position to extension and neutral (Fig. 82.5A–C). The examiner's hand should be placed on the groin to palpate the iliopsoas snapping, which typically occurs with the hip between 30 and 45 degrees of flexion.[51] Iliopsoas strength is assessed by resisted hip flexion with the patient in the sitting position, which may result in groin pain but does not usually recreate snapping. Localized swelling of the inguinal region has been reported in up to 59% of patients with painful internal snapping.[27] Diagnostic US-guided injections of the iliopsoas bursa are useful in the evaluation of iliopsoas snapping (Fig. 82.6).[52] A pre- and postinjection examination can be performed to determine whether the patient experiences pain relief; if so, it supports the diagnosis of painful snapping.[53]

It is important also to evaluate patients for external snapping of the iliotibial band (ITB) over the greater trochanter, which may present in a similar manner to iliopsoas snapping. Patients often report a sensation of the hip dislocating during the snapping event. The examination is the most efficient way to distinguish between internal and external hip snapping. Our preferred exam technique—the "bicycle test"—for determining the presence of external hip snapping is performed by having the patient actively cycle the affected extremity from flexion to extension while lying in the lateral decubitus position (Fig. 82.7A and B). A palpable snap or clunk over the greater trochanter is confirmatory for this diagnosis.

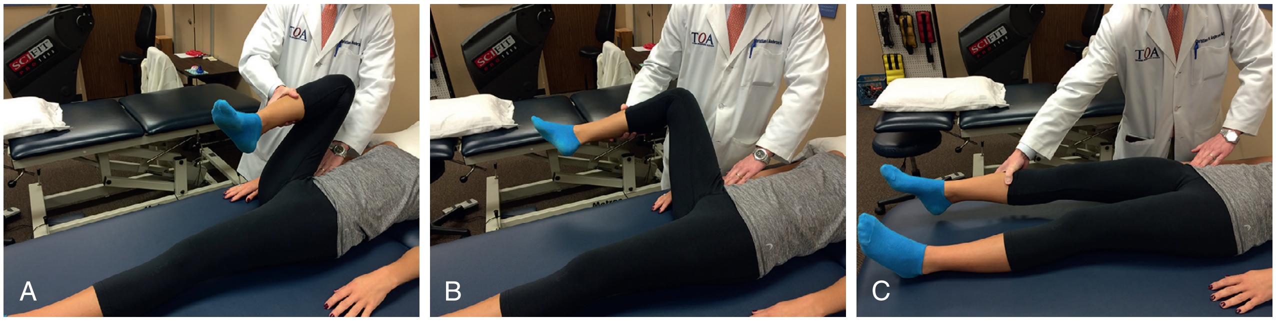

Fig. 82.5 The "active iliopsoas snapping test" for internal snapping of the iliopsoas. The patient actively moves the hip from flexion (A) to abduction and external rotation (B) and then to extension and neutral (C). A palpable clunk or pop is often felt with the examiner's hand placed over the hip.

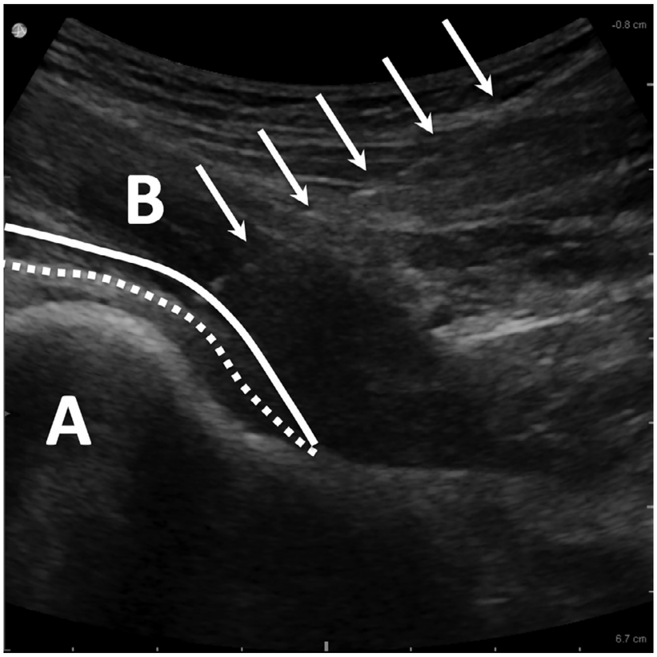

Fig. 82.6 Ultrasound-guided injection of the iliopsoas bursa. The needle trajectory *(arrows)* is toward the femoral head (A). The tip of the needle should penetrate through the iliopsoas muscle (B) into the iliopsoas bursa, which is located between the posterior surface of the iliopsoas *(solid line)* and the joint capsule *(dashed line)*.

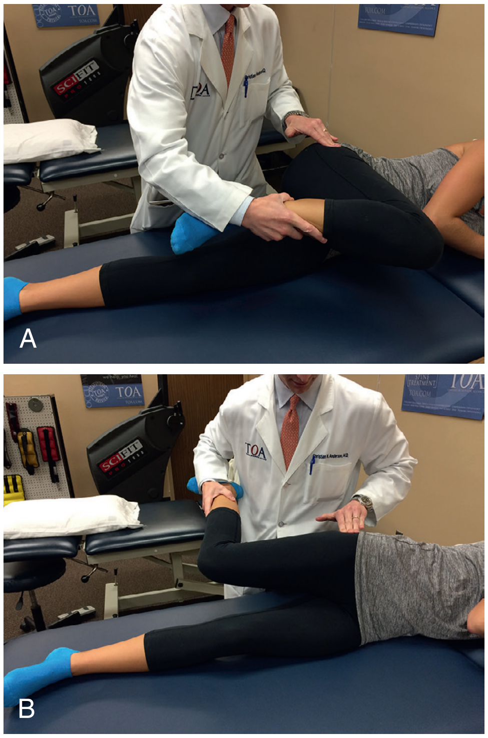

Fig. 82.7 The "bicycle test" for external snapping of the iliotibial band. In the lateral decubitus position, the patient actively flexes (A) and then extends (B) the hip. A palpable, audible, and/or visible clunk can be detected over the greater trochanter.

Iliopsoas Impingement

Patients with iliopsoas impingement typically have a positive impingement test (flexion, adduction, and internal rotation), scour sign (flexion, adduction, and axial compression), and tenderness with manual compression over the iliopsoas.[40,48,49] Approximately half of patients have pain with FABER and resisted straight leg raise testing.[48,49] Intra-articular injections have shown variable results, with some studies reporting transient improvement in 50% of patients[40] while others report improved symptoms in all patients undergoing injection.[48]

IMAGING

Although iliopsoas snapping is typically diagnosed with a thorough history and physical exam, imaging studies can be valuable for confirming the diagnosis and identifying concomitant hip pathology. Radiographic evaluation should begin with anteroposterior (AP) radiographs of the pelvis and lateral hip to rule out acute or chronic osseous abnormalities and evaluate for radiographic signs of FAI. In cases where the source of snapping is uncertain, dynamic US can be used to visualize the iliopsoas tendon or ITB during provocative maneuvers such as the active iliopsoas snapping and bicycle tests.[9,24–29,34] US is also useful in identifying joint effusions and synovitis, rectus femoris tendinopathy, and iliopsoas bursitis and tendonitis.[27,34] In addition to also being able to detect iliopsoas tendinitis and bursitis,[25] MRI is useful in diagnosing associated chondral and labral pathology, which is present in 67% to 100% of patients presenting

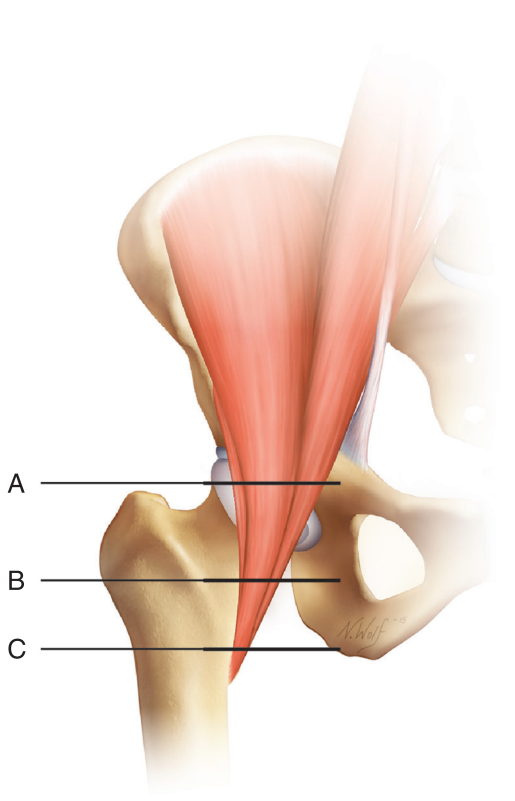

Fig. 82.8 The three described levels of iliopsoas release: central compartment *(A)*, peripheral compartment *(B)*, and lesser trochanter *(C)*. At these levels, the ratio of tendon to muscle is 40% tendon/60% muscle belly, 53% tendon/47% muscle belly, and 60% tendon/40% muscle belly, respectively.[71] (Copyright 2017, Nicole Wolf, all rights reserved.)

with painful iliopsoas snapping.[53–56] MRI can determine the pathologic reason for snapping in up to 100% of patients.[27] For patients with suspected IPI, plain film x-rays may show signs of FAI[49]; however, the most pertinent radiographic finding is a labral tear at or near the 3 o'clock position as seen on MRI.[48]

TREATMENT OPTIONS

The first line of treatment for both iliopsoas snapping and impingement is a conservative program including activity modification, physical therapy, nonsteroidal antiinflammatory drugs (NSAIDs), and corticosteroid injections. Surgical treatment is considered in patients who have failed a 3-month conservative course. The goal of surgical treatment is to lengthen the iliopsoas musculotendinous unit to prevent snapping or mechanical overpressurization of the underlying labrum. Iliopsoas lengthening can be performed by either an open or an arthroscopic approach. Arthroscopic release of the iliopsoas tendon has been described at three locations (Fig. 82.8)—in the central compartment (Fig. 82.9),[49,54,55,57] in the peripheral compartment (Fig. 82.10),[58,59] and at the lesser trochanter (Fig. 82.11).[53,56,60,61] Labral pathology associated with FAI or iliopsoas impingement is typically treated with correction of bony abnormalities followed by labral repair or débridement as indicated by tissue quality at the time of surgery.

POSTOPERATIVE MANAGEMENT

The postoperative rehabilitation program begins immediately after surgery and is a multiphase program that lasts for 25 weeks. The goals of therapy are to protect tissue repairs, preserve ROM, decrease inflammation and pain, and restore normal muscle strength and function. Phase I (0 to 2 weeks) consists of initial exercises, and patients are restricted to 20 lb foot-flat weight bearing with crutches. Passive ROM and stretching exercises are emphasized, with hip flexor stretching and hip extension limited to the neutral position to protect capsular repairs. Patients also begin active ROM, isometric strengthening, stationary bike with no resistance, and early proprioceptive exercises. Patients are advised to take naproxen 500 mg twice daily for 3 weeks to decrease the likelihood of heterotopic ossification and control inflammation during early rehabilitation.[64] In phase II (3 to 12 weeks), patients are weaned off crutches and advanced to intermediate exercises. Hip flexor exercises are started after 6 weeks to allow healing of the released iliopsoas tendon. Exercises include progression of balance and proprioceptive exercises, cardiovascular exercises (stationary bike with resistance and elliptical), and lower extremity strengthening. Phase III (13 to 16 weeks) consists of advanced exercises including plyometric progression, running, and agility drills. Last, patients are advanced to phase IV (17 to 25 weeks), which includes high-level activities such as multiplane agility and sport-specific drills.

RESULTS

Iliopsoas Snapping

Historically an open surgical approach was used for iliopsoas release; however, open surgery has been shown to be associated with increased morbidity and inferior results compared with arthroscopic techniques. In a systematic review of 11 studies, Khan et al.[65] found that patients undergoing open procedures had more postoperative pain, and recurrent snapping occurred in 23% of open compared to 0% of the arthroscopic surgeries. Furthermore, patients who underwent open surgery had a complication rate of 21% compared with 2.3% using arthroscopic techniques.[65] In addition to the lower rate of complications and recurrent snapping, the arthroscopic approach can be used to diagnose and treat concomitant intra-articular pathology. Treatment of associated intra-articular abnormalities in conjunction with iliopsoas snapping, relative to open procedures, may result in improved patient-reported outcomes. Currently, however, there are no direct comparative studies evaluating open versus arthroscopic techniques for iliopsoas lengthening.

Case series studies (LOE IV) evaluating arthroscopic iliopsoas release universally report good to excellent patient reported outcomes (PROs), low recurrence rates, and minimal complications.[53–59] In a study (LOE IV) evaluating athletes with painful iliopsoas snapping, Anderson and Keene[53] reported return to sport in all patients at an average of 9 months after release of the tendon at the lesser trochanter. In a randomized trial (LOE I) and a comparative study (LOE IV), Ilizaliturri and colleagues[66,67] found favorable results with iliopsoas release at either the lesser

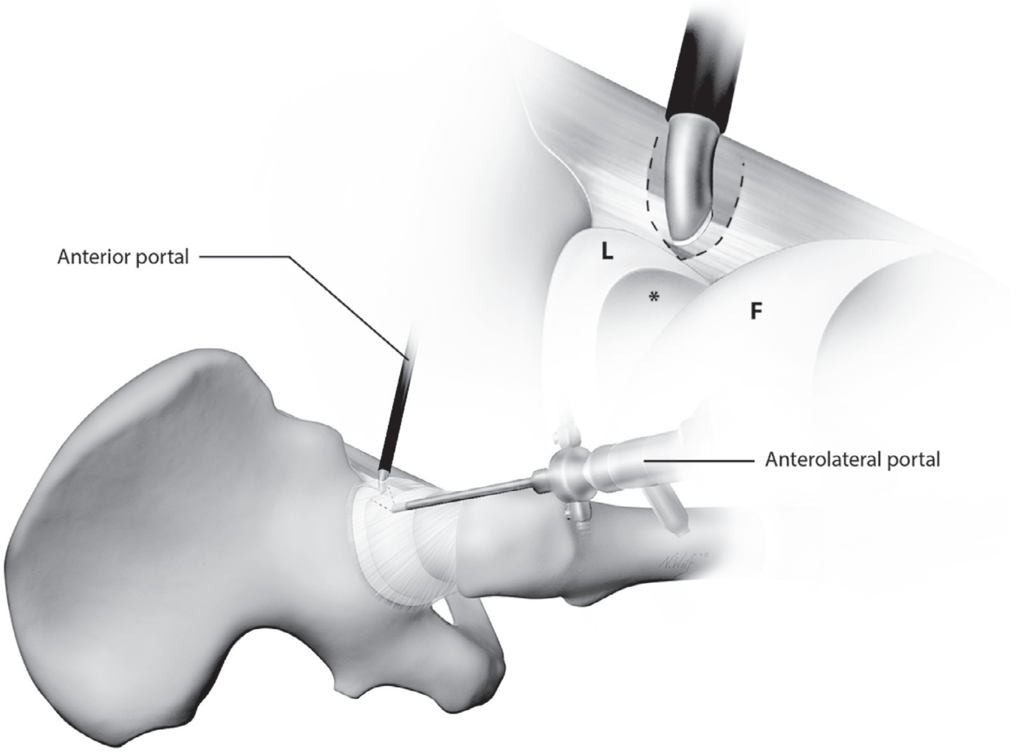

Fig. 82.9 Release of the iliopsoas tendon in the central compartment. With the hip under traction and a 70-degree arthroscope in the anterolateral portal, a banana blade is used to extend the interportal capsulotomy medially *(dashed line)* to expose the iliopsoas tendon, located just anterior to the iliopsoas notch at the 3 o'clock position on the acetabulum *(*)*. The tendon can then be released with the blade or an electrocautery device, taking care to leave the muscular portion of iliopsoas intact. This results in a fractional lengthening of the musculotendinous unit. *F*, Femoral head; *L*, labrum. (Copyright 2017, Nicole Wolf, all rights reserved.)

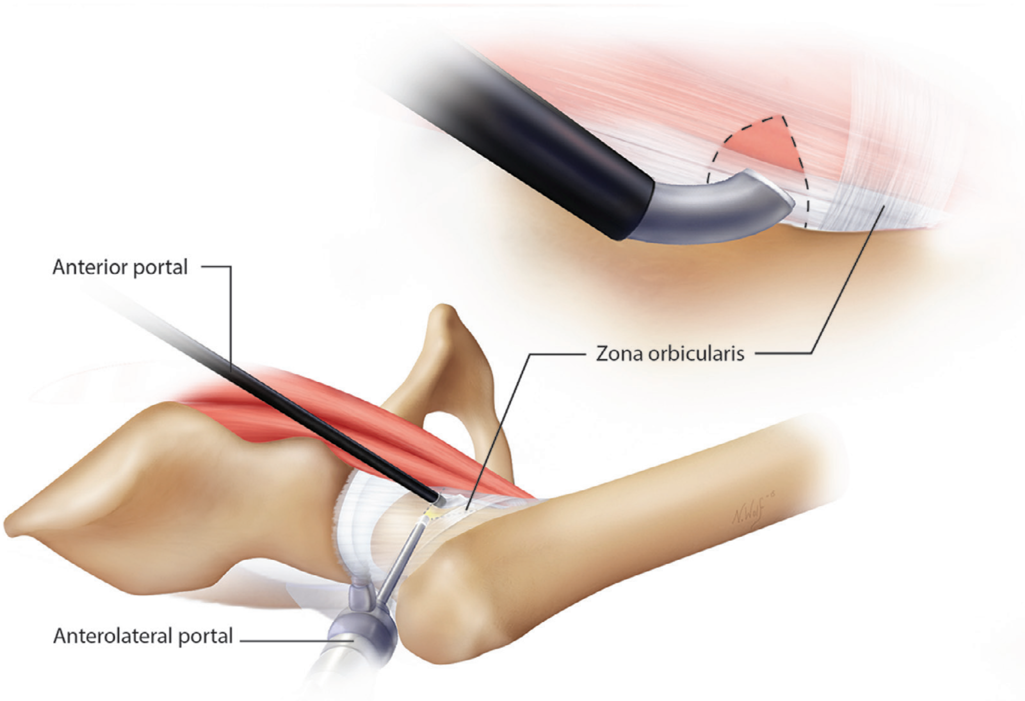

Fig. 82.10 Iliopsoas tenotomy in the peripheral compartment. To facilitate visualization, traction is released and the hip is placed in 30 degrees of flexion. A 30-degree arthroscope is placed in anterolateral portal and pointed anteriorly towards the joint capsule. A small (1 cm) transverse capsulotomy is made lateral to the medial synovial fold and just proximal to the zona orbicularis anteriorly. The tendon is identified through the capsulotomy and then divided with an electrocautery device or banana blade. (Copyright 2017, Nicole Wolf, all rights reserved.)

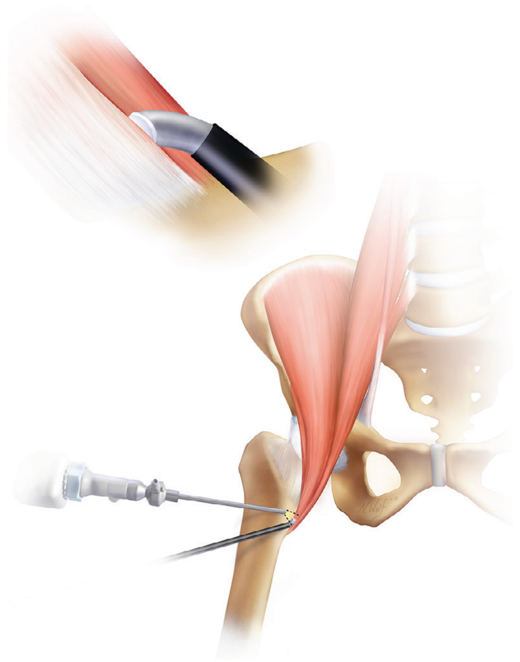

Fig. 82.11 Iliopsoas tendon release at the lesser trochanter. To access the lesser trochanter, the hip is flexed 30 degrees and externally rotated until the lesser trochanter is parallel to the coronal plane of the body and maximally visualized with fluoroscopy. A spinal needle is then advanced anterior and perpendicular to the femur until it reaches the lesser trochanter. A cannula is placed in this position and a second portal is made in a similar manner 5 to 7 cm distal to the first one. The 30-degree arthroscope is then placed in the proximal portal and an electrocautery is used in the distal portal to clear any soft tissues and release the iliopsoas tendon at its insertion on the lesser trochanter. (Copyright 2017, Nicole Wolf, all rights reserved.)

trochanter or in the central compartment, with no significant differences between the techniques. Fabricant and colleagues (LOE IV)[57] studied 67 consecutive patients undergoing iliopsoas release in the central compartment and determined that patients with high femoral anteversion (>25 degrees) had lower modified Harris hip scores (HHS) compared with patients who had normal/low (≤25 degrees) femoral anteversion. These investigators hypothesized that the iliopsoas may act as an important dynamic stabilizer given its anatomic location at the anterior aspect of the hip joint.[57] Even so, Chandrasekaran et al.[68] (LOE IV) demonstrated that although females had higher mean rates of femoral anteversion compared with male patients, they had greater improvements in their PROs. These investigators suggested that with appropriate restoration of other soft tissue constraints, such as the labrum and capsule, fractional lengthening of the iliopsoas can be performed without compromising outcomes.[68] Overall, additional research is necessary to determine predictors of functional and patient-reported outcomes and the best technique of iliopsoas release.

Iliopsoas Impingement

Surgical management of IPI with iliopsoas lengthening and treatment of concurrent labral pathology has demonstrated favorable results in LOE IV studies.[40,49,50] Domb and colleagues,[40] demonstrated that 95% of patients surveyed reported that their physical ability was "much improved" and none reported worse symptoms at an average of 21 months after arthroscopic tendon lengthening and either labral débridement or repair. HHS and hip outcome scores (HOS) were available in eight patients at final follow-up; these demonstrated significant improvements compared with preoperative scores.

Author's Preferred Technique

Our preferred surgical technique for the treatment of iliopsoas disorders is an arthroscopic release in the central compartment. We typically correct bony abnormalities associated with FAI and treat labral pathology in the central compartment prior to performing release of the iliopsoas tendon to decrease the chance of intra-abdominal fluid extravasation.[62,63] The patient is positioned supine on the operating table against a well-padded perineal post. Both feet are placed in padded traction boots and the operative extremity is positioned in 10 degrees of flexion, 10 degrees of abduction, and neutral rotation. The contralateral extremity is placed in 45 to 60 degrees of abduction, neutral rotation, and neutral flexion and serves as countertraction to the operative leg. The C-arm is placed between the lower extremities. Traction is then incrementally applied to the operative leg and fluoroscopy is used to confirm that at least 1 cm of hip joint distraction has been obtained. Paralysis is used to allow adequate distraction of the hip during traction.

We prefer the use of three portals for access to the central compartment—the anterolateral, midanterior, and posterolateral portals (Fig. 82.12). The anterolateral portal is placed 1 cm anterior to the tip of the greater trochanter using the Seldinger technique. A 17-gauge spinal needle is introduced first, and an air arthrogram is performed to ensure that the needle is within the central compartment. Once central compartment access has been confirmed, a nitinol wire is introduced through the spinal needle and a cannulated trocar is then placed over the guidewire into the central compartment. A 70-degree arthroscope is introduced into the hip and the pump is not turned on until the midanterior portal has been established. The midanterior portal is placed approximately 2 cm lateral and 2 cm distal to the junctional point formed between a horizontal line drawn medially from the tip of the greater trochanter and a vertical line drawn distally from the anterosuperior iliac spine (ASIS) (see Fig. 82.12). Viewing anteriorly with the arthroscope in the anterolateral portal, the midanterior portal is established under direct visualization by placing the spinal needle through the capsule midway between the labrum and the femoral head. The pump is then turned on and the camera is directed posteriorly to establish the posterolateral portal, which is placed 1 to 2 cm posterior to the greater trochanter. The arthroscope is then switched to the posterolateral portal and viewing anteriorly. A banana blade is used to make an interportal capsulotomy between the anterior and anterolateral portals. For patients with suspected iliopsoas impingement, the posterior portal allows excellent visual inspection of the labrum and chondrolabral junction at the psoas-U (see Fig. 82.4). A probe is introduced through the midanterior and anterolateral portals to allow complete inspection of the labrum and articular surface of the acetabulum. The camera is then switched to the anterolateral portal and, using the banana blade in the midanterior portal, the capsulotomy is extended medially toward the psoas-U. The blade is used with a controlled sawing motion, taking care to section only the joint capsule. It is important to note that the capsulotomy should be performed so that there is adequate tissue on the proximal and distal edges of the capsule to allow a repair at the end of

Author's Preferred Technique—cont'd

the procedure. The psoas tendon is visualized by its distinct glistening fibers and is located immediately anterior to the capsule at the 3 o'clock position on the acetabular clock face. A shaver is then used to resect the tenosynovium circumferentially from around the tendon. The tendon is then bisected using an electrocautery device or banana blade, leaving the muscle fibers intact. Since anatomic studies have shown that up to 72% of individuals may have more than one tendon slip,[6] we routinely search for additional tendons. If a smaller iliacus tendon is encountered first, the larger psoas tendon will be found medial to the iliacus tendon and should be sectioned in a similar manner. Once the tendons have been lengthened, the traction can be released and pathology in the peripheral compartment can be addressed. We routinely perform a capsular repair in the setting of iliopsoas release to minimize the chance of iatrogenic instability.

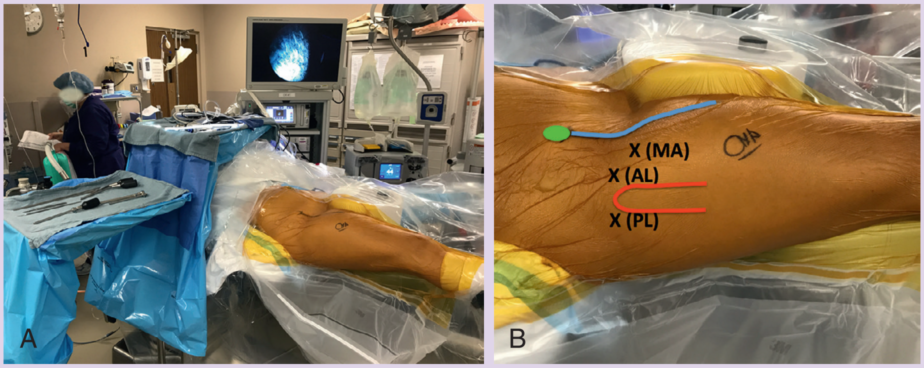

Fig. 82.12 (A) The operating room setup showing (B) the location of the arthroscopy portals. *AL*, Anterolateral; *ASIS*, anterior superior iliac spine; *blue line*, line drawn from the ASIS distally toward the center of the patella; *green oval*, ASIS; *MA*, midanterior; *PL*, posterolateral; *red line*, outline of greater trochanter.

In a case series of 16 patients with a minimum of 6 months of follow-up, Cascio and coworkers[50] demonstrated improvement in HHSs from a mean of 70 to 94 after tendon lengthening with or without labral repair. The authors noted, however, that one patient required revision surgery at 18 months for repair of a labral tear that was not addressed at the initial surgery. Nelson and Keene[49] reported good to excellent results (modified HHS ≥80 points) in 23 of 30 patients undergoing tendon release for IPI. Patients with lower final PROs had avascular necrosis ($n = 1$), progressive degenerative joint disease ($n = 1$), trochanteric bursitis ($n = 2$), or recurrent painful iliopsoas snapping ($n = 3$). Two of the three patients with recurrent symptomatic snapping underwent a second iliopsoas release at the lesser trochanter and subsequently demonstrated good to excellent outcomes 1 year after the revision surgery. Although these Level IV reports are encouraging, further studies with larger sample sizes and long-term follow-up are necessary to determine the optimal treatment for IPI.

COMPLICATIONS

Although the outcomes of iliopsoas release are generally good, several complications specific to this procedure have been reported. Postoperative hip flexor weakness[53,55,56,58,59,61,67,69] and atrophy of the iliopsoas on MRI[69,70] have been observed after tenotomy. There is consensus in the literature that significant weakness occurs in the early postoperative period but also controversy regarding whether this weakness resolves over time. Several studies evaluating posttenotomy iliopsoas strength using a standard 5-point manual muscle testing scale have demonstrated complete resolution of the weakness by 3 to 6 months after surgery.[53,55,56] Furthermore, studies by Ilizaliturri et al.[61,67] and Hwang et al.[58] demonstrated "improvement" or "recovery" of strength by 8 to 10 weeks; however, no formal strength testing was performed. In contrast, Brandenburg and associates [69] (LOE III) used a dynamometer mounted to a custom test frame to demonstrate a 19% decrease in seated hip flexion strength and noted a 25% decrease in the iliopsoas volume measured on MRI reconstructions compared with the contralateral hip at a mean of 21 months post surgery. The reason for the discrepancy in postsurgical hip flexor strength between these studies is unclear but may involve the differing techniques of measuring strength (hand measurements vs. dynamometer). Interestingly, no significant differences in postoperative hip flexion strength have been observed with tendon release at the different described levels.[65–67] The latter observation can partly be explained by the similar ratio of muscle-to-tendon volume observed at these locations.[71] Consequently each technique results in a comparable volume of muscle fibers remaining intact within the musculotendinous unit, allowing similar forces to be generated for hip flexion.

Other reported complications after iliopsoas release include recurrent snapping,[17,54] fluid extravasation,[62,63,72] and gross hip instability.[73–75] Shu and Safran[17] (LOE V) reported a case of recurrent snapping due to a bifid iliopsoas tendon that was not recognized during release in the peripheral compartment. The recurrent snapping resolved after revision arthroscopy and release of the two tendinous slips. In a Level IV case series, Bitar and coworkers[54] noted that 18% of patients developed recurrent snapping after iliopsoas lengthening in the central compartment. Patients with recurrent snapping had no improvement in PROs and had lower satisfaction compared with those whose snapping had resolved.[54] Although the exact causes of recurrent snapping were not noted, the authors hypothesized that multiple tendon slips, scar tissue formation, or tightness in the muscular portion of the iliopsoas may have contributed.[54] These studies underscore the importance of assessing both preoperative MRIs and intraoperatively for the presence of multiple tendons.

Fluid extravasation is a rare but potentially life-threatening complication that can result in abdominal compartment syndrome.[72] A recent systematic review (LOE IV) of 14 studies by Ekhtiari and coworkers[72] demonstrated that the prevalence of symptomatic abdominal extravasation was 1.6%. They concluded that female sex was the only variable found to increase the risk of fluid extravasation and that iliopsoas tenotomy was not a risk factor.[72] However, in a study evaluating postarthroscopy CT scans, Hinzpeter et al.[62] (LOE IV) demonstrated intra- or retroperitoneal fluid extravasation in 47.5% of asymptomatic patients. In this study, iliopsoas tenotomy was associated with a nonstatistical increase in the mean volume of fluid extravasation.[62] Other risk factors for increasing extravasation were female sex and operative time.[62] Kocher and colleagues[63] performed a survey (LOE IV) for the Multicenter Arthroscopy of the Hip Outcomes Research Network (MAHORN) and determined that the prevalence of symptomatic intra-abdominal fluid extravasation was only 0.16%. In agreement with the study by Hinzpeter, they determined that along with fluid pump pressure, iliopsoas release was a risk factor for developing extravasation.[63] The authors determined that this complication was likely the result of fluid tracking along the iliopsoas sheath following iliopsoas tenotomy.[63] Consequently they recommended performing tendon release at the end of the procedure.[63] Symptoms of intra-abdominal fluid extravasation include abdominal pain, hypotension, hypothermia, abdominal distention, shortness of breath, and cardiopulmonary arrest.[63,76] Using clear arthroscopy drapes (Fig. 82.12), frequently palpating the abdomen, and monitoring the core body temperature are recommended to assist in the earlier detection of this complication.[72] If intra-abdominal extravasation is identified, immediate cessation of the procedure, diuresis, warming of the core temperature, and general surgery consultation are warranted.[72]

Gross hip instability is another rare but potentially devastating complication that has been reported after iliopsoas release.[73–75] A systematic review (LOE IV) by Duplantier and associates[73] determined that 3 of 11 reported cases of hip dislocation or subluxation reported in the literature after hip arthroscopy were associated with iliopsoas release. A labral débridement was performed in one of the three cases,[74] and capsular closures were not reported in any of them.[74,75] The iliopsoas has been hypothesized to function as a dynamic anterior stabilizer of the hip.[43,57] However, it is unclear what its overall contribution to hip stability is relative to the bony anatomy of the acetabulum and other soft tissue stabilizers, such as the capsule, labrum, and ligamentum teres. Given the multiple confounding variables associated with gross hip instability after arthroscopy, further research is needed to determine the role of iliopsoas release in the development of this complication.

FUTURE CONSIDERATIONS

The iliopsoas is an anatomically complex musculotendinous unit that functions primarily as a powerful hip flexor and secondarily as a femoral rotator and stabilizer of the lumbar spine and pelvis. Initial treatment of iliopsoas disorders generally consists of a combination of physical therapy, activity modification, NSAIDs, and corticosteroid injections. If conservative treatment fails, arthroscopic surgery to address the existing pathologic condition has demonstrated encouraging results mostly in Level IV studies. Further studies with higher levels of evidence and longer-term follow-up and studies utilizing validated strength measurement testing and PROs are needed to determine both positive and negative predictors of outcomes. Biomechanical studies are also needed to determine the exact role of the iliopsoas in acting as a dynamic stabilizer to the hip joint.

For a complete list of references, go to ExpertConsult.com.

SELECTED READINGS

Citation:
Philippon MJ, Devitt BM, Campbell KJ, et al. Anatomic variance of the iliopsoas tendon. *Am J Sports Med*. 2014;42(4):807–811.

Level of Evidence:
Descriptive laboratory study

Summary:
This article is an anatomic study of 53 cadavers in which single-, double-, and triple-bundled iliopsoas tendons were noted in 28%, 64%, and 8% of patients, respectively.

Citation:
Deslandes M, Guillin R, Cardinal E. The snapping iliopsoas tendon: new mechanisms using dynamic sonography. *AJR Am J Roentgenol*. 2008;190(3):576–581.

Level of Evidence:
Descriptive imaging study

Summary:
Dynamic US was used in this study to demonstrate that iliopsoas snapping occurs when the iliacus muscle becomes trapped between the iliopsoas tendon and the superior pubic ramus, then suddenly releasing as the hip is brought from FABER to neutral, resulting in the tendon snapping against the bony pelvis.

Citation:
Domb BG, Shindle MK, McArthur B, et al. Iliopsoas impingement: a newly identified cause of labral pathology in the hip. *HSS J*. 2011;7(2):145–150.

Level of Evidence:
IV

Summary:
This article defined the pathophysiology of iliopsoas impingement (IPI), determining that most individuals with IPI had labral tears or bruising at the 3 o'clock position on the acetabulum. Iliopsoas release resulted in decreased compression of the iliopsoas on the underlying labrum and improvements in validated outcome scores.

Citation:
Anderson SA, Keene JS. Results of arthroscopic iliopsoas tendon release in competitive and recreational athletes. *Am J Sports Med*. 2008;36(12):2363–2371.

Level of Evidence:
IV

Summary:
This study evaluated iliopsoas release at the lesser trochanter in 15 athletes with a history of painful iliopsoas snapping. Significant improvements were observed in mHHS and all patients returned to sport within 9 months after surgery.

Citation:
Bitar El YF, Stake CE, Dunne KF, et al. Arthroscopic iliopsoas fractional lengthening for internal snapping of the hip: clinical outcomes with a minimum 2-year follow-up. *Am J Sports Med*. 2014;42(7):1696–1703.

Level of Evidence:
IV

Summary:
This study evaluated 55 patients who underwent iliopsoas release in the central compartment and demonstrated that the majority of patients had resolution of snapping and improvement in validated outcome scores. Patients who did not have resolution of snapping (18%) had no significant improvement in outcomes.

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